Microloans, Macro Patterns: Uncovering Trends in Kiva Borrowing

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Abstract

In the absence of adequate social services and economic opportunities in many developing countries, microloans remain a vital source of capital injection necessary for individuals to provide for their families and support their local communities. Hence, it becomes critical to understand how different factors, such as gender or the location of borrowers, correspond to the sectors of the local economy to which the loans are put to use.

To investigate these patterns, we analyzed Kiva loans using two algorithms: conditional inference trees (Algorithm 1) and random forests (Algorithm 2). Across both algorithms, we found evidence that gender, term, and continent were the strongest predictors of loan use for each year of data (2014-2017). Furthermore, loans with terms longer than 14 months were most likely to be within the agricultural sector, regardless of other factors, while shorter loans were more likely to differ based on gender and then subsequently also by continent. Contrary to our expectations, we had a disproportionate amount of borrowers across different sectors (see Appendix D), resulting in similar misclassification rates around 65-67% for both algorithms. However, the similarity in misclassification rates between our training and testing data subsets indicates no overfitting. Overall, our findings highlight recurring patterns in how different demographic and geographic groups utilize microloans, providing preliminary guidance for lenders seeking to better target resources to sectors and communities most dependent on microloans.

1. Introduction

According to the American Foreign Service Association, an estimated 1.7 billion adults around the world remain unbanked, comprising an enormous pool of potential recipients of microloans provided by NGOs and various foundations (Dichter). One of the biggest drivers, Kiva, has already lent over $1.6 billion to more than 4.1 million borrowers in more than 144 countries (Leigh Lebos). This rise in recipients fuels the need for Kiva and its partner organizations, local lending subgroups, and global organizations to understand the existing record of microloan use and the insights their use might reveal about the communities utilizing them. To this end, Kiva has released a substantial amount of data on over 600,000 microloans through Kaggle in 2017, a crowdsourced data science platform, allowing users to help Kiva estimate levels of poverty in the communities using information about the loans, such as lender region and loan amounts (Collins). However, analyses of the data set we were able to find do not address our specific area of interest regarding loan use based on a subset of specific factors.

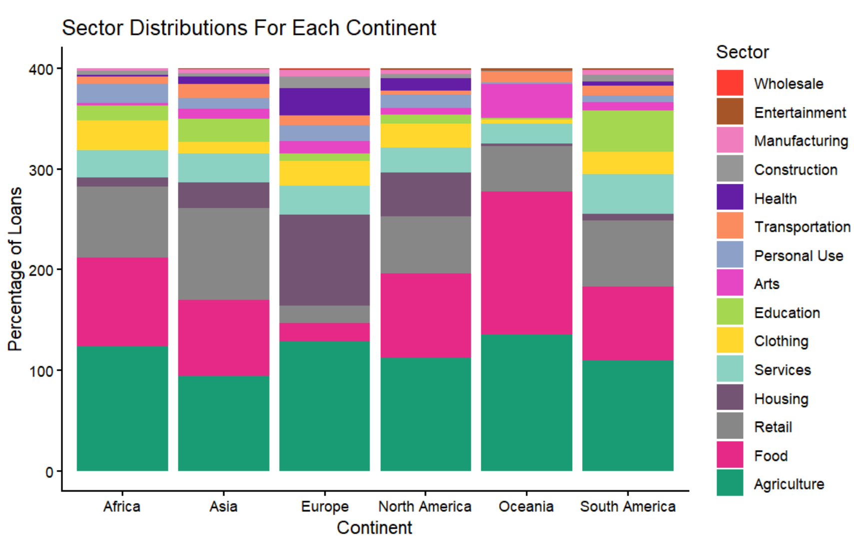
We frame our investigation around the following question: What are the important predictors of loan use among Kiva recipients? For our investigation, we define loan use as the variable *sector*. Specifically, there are 13 total sectors: food, agriculture, wholesale, construction, arts, transportation, services, manufacturing, retail, clothing, health, education, personal use, housing, and entertainment. By uncovering predictable patterns in sectors, we hope to emphasize to policymakers which sectors of the economy would benefit most from future economic reforms based on their dependency on microloans.

2. Data and Methods

a. Dataset Description and Exploration

For our analysis, we used the original dataset provided by Kiva that included 671,205 loans from 144 countries between 2014 and 2017. While the original dataset had 20 columns describing each loan, we thoroughly cleaned it to prepare the data for the modeling tree algorithms. As such, we kept variables like *sector*, *borrower gender*, limited to individual loans rather than groups, *term in months* to repay the loan, *country*, which we condensed to *continent*, as well as *date,* which we thought would be most relevant for our analysis and proved most promising in our preliminary analysis. We also anticipated that year and continent may differ in the sectors the loans are being applied for, thus we grouped the loans by year and continent.

In our exploration, the differences in sector distributions across the years differed slightly, but not by much (Appendix A). In contrast, we saw some variations in sector use by continent. Continents with Western Countries– specifically Europe and North America– had larger proportion of loans dedicated to



Housing. Across all continents, however, the largest majority loans were dedicated to *Agriculture, Food, and Retail*.

Figure 1: Proportion Distribution of Sectors within Each Continent

b. Statistical Analysis Framework

Algorithm 1: After dropping missing rows, we had 561,212 rows in our dataset, from which we randomly took 107,500 to sample due to the massive size of our dataset. We allocated 80% of the sample data for training, and the remaining 20% for testing. We applied a stratified sampling technique to the data, randomly selecting 30,000 rows of loans from each year. Algorithm 1 constructs splits based on conditional inference theory, testing all possible splits among the different variables that lead to the most distinct predictions in *sector*. By choosing to use a conditional inference tree, we ensure our results are easily accessible and allow for determining patterns in data.

Algorithm 2: Drawing on the subset of data, we similarly use random forests to model a random forest of the subset. By taking random subsets of the data using bootstrapping and random features, the algorithm creates a set number of trees (600 in this case) each time a random forest is run. The algorithm splits based on how mixed the sectors are in a particular tree node (Gini impurity), or on the residual sum of squares (RSS). This is to ensure that the predictions take into account the uncertainty that may be present with the loans themselves.

3. Results

After using our sample dataset and inputting it into the two packages we have tested, we have created several trees and other visualizations to showcase our results. Using algorithm 1, we found that our results differed by year, and so we were able to create visualizations like Fig. 2 as shown in detail in the appendix. From those graphs, we were able to summarize our tree results in Fig. 3 below, which lists the most important variables that the different packages deemed, with the randomForest package using the mean decrease accuracy measure.

	Predictor	2014		2015		2016		2017	
Algorithm 1 & **Algorithm 2**	gender	1 ( )	**1**	3	**2**	3	**1**	3	**1**
	term	2	**2**	1	**3**	1	**2**	1	**2**
	continent	3	**3**	2	**1**	2	**3**	2	**3**
Algorithm 2	month	**4**		**4**	**4**[1]	**4**		**4**	

Figure 3: Variables Most to Least Important for Predicting Sectors According to Algorithms 1&2 through 2014-2017

For 2014, the most important predictor rankings were the same across both Algorithm 1 and Algorithm 2; hence, we know that the order of importance is the most accurate. However, for the other years, they were different, indicating that the robustness of the packages comes into play. Specifically, while random forest modeling in Algorithm 2 was likely highly affected by the class imbalance from our sector variable, this might not have been so for Algorithm 1. Instead, Algorithm 1 likely removed highly predicted calculators while Algorithm 2 might not have, hence their threshold for correlation might be different.

To evaluate our models, we calculated the misclassification rate using each algorithm. Although the numbers were high, there wasn't much difference between the training and testing, which indicates that our training model doesn't overfit our data.

[1] Month only appeared once in the variable importance models from Algorithm 1 for 2015.

	2014		2015		2016		2017	
Training Data	0.674	**0.644**	0.675	**0.644**	0.675	**0.644**	0.674	**0.644**
Testing Data	0.675	**0.654**	0.673	**0.654**	0.672	**0.656**	0.675	**0.656**

Figure 4: Misclassification Rates Amongst Sectors Across Training and Testing Data Algorithm 1 and Algorithm 2 (bolded) from 2014-2017

Most often, *Food* was misclassified as retail, while *Retail* was also misclassified as *Agriculture*, and *Agriculture* was often misclassified as *Food.*

4. Conclusion

a. Answering Research Question

In conclusion, we have determined that the most important predictors of loan use amongst the Kiva loans, except for 2015, from 2014-2017 were gender, term, and continent in that order. Looking at the trees produced by both algorithms, they all split on the same cutoff and level of predictors. First by term, either being greater than or less than/equal to 14 months, then by gender. For the female split, the trees then split again by continent – Asia and Oceania in one split and the other continents in the other, (see appendix section parts B and C). The Algorithm 1 model has higher misclassification rates than the Algorithm 2 model, but both models are comparable and do not overfit. Our analysis shows that sectors receiving microloans are most strongly influenced by term, gender, and geographic region. By identifying these drivers, we provide evidence that agricultural and service-based sectors are the most dependent on microloans and therefore the most likely to benefit from targeted economic reforms.

b. Limitations

We acknowledge there are several limitations to our data analysis and subsequent use. For example, whilst using the misclassification rate as a metric for evaluating our models, we did not account for class imbalances within *sector*. Hence, some sectors had more data than others for the model to estimate its predictions from. We also had significant computing difficulties, as the original dataset was too large; hence, we used a much smaller sample to run our models. Our dataset is also from 8 to 11 years ago. Hence, even though our model is able to predict sectors based on the variables in our sample, we recommend that relevant organizations use more recent Kiva loan data for the most accurate predictions in the sector. Given what we also know about the continent distribution, Future iterations of our model would look to run our models by continent rather than year.

During cleaning, we also dropped several variables with either missing values, which correlated with another variable, or which did not prove relevant from our tree algorithms. For example, the region was correlated with the continent; meanwhile, the borrower ID was not relevant. Having dropped variables could potentially result in introducing bias in our results if the data is not missing completely at random, giving us less generalizable results and weaker conclusions.

Works Cited

Collins, Elliot. "Data Science for Good: Kiva Crowdfunding." *Kaggle*, 2017, www.kaggle.com/datasets/kiva/data-science-for-good-kiva-crowdfunding/discussion/54320.

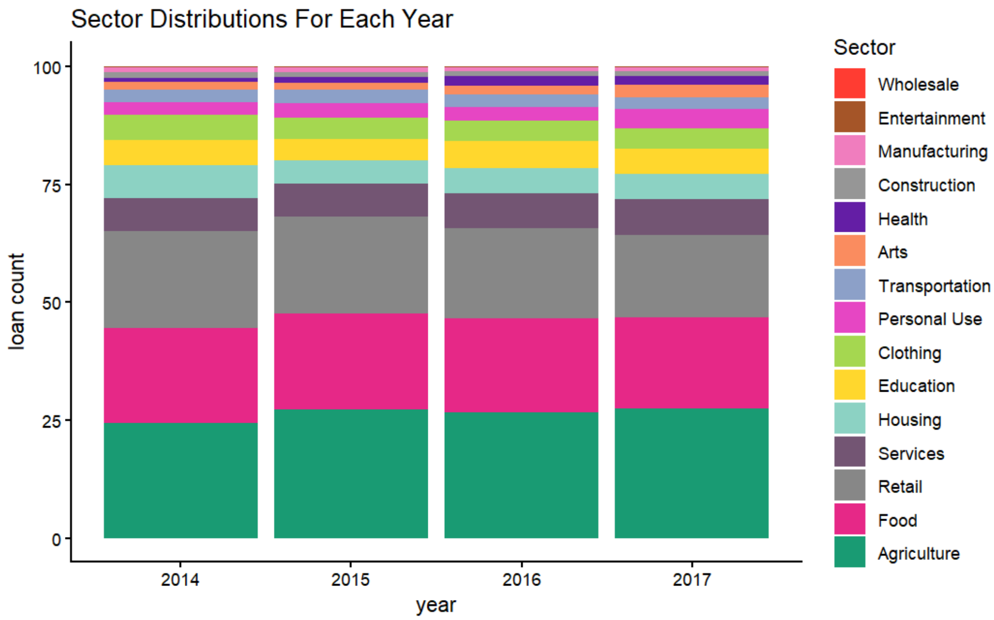
DICHTER, THOMAS. "Whatever Happened to Microfinance? A Cautionary Tale." *Afsa.org*, 2023, afsa.org/whatever-happened-microfinance-cautionary-tale.

Donou-Adonsou, Ficawoyi, and Kevin Sylwester. "Growth Effect of Banks and Microfinance: Evidence from Developing Countries." *The Quarterly Review of Economics and Finance*, vol. 64, May 2017, pp. 44–56, https://doi.org/10.1016/j.qref.2016.11.001. Accessed 30 July 2020.

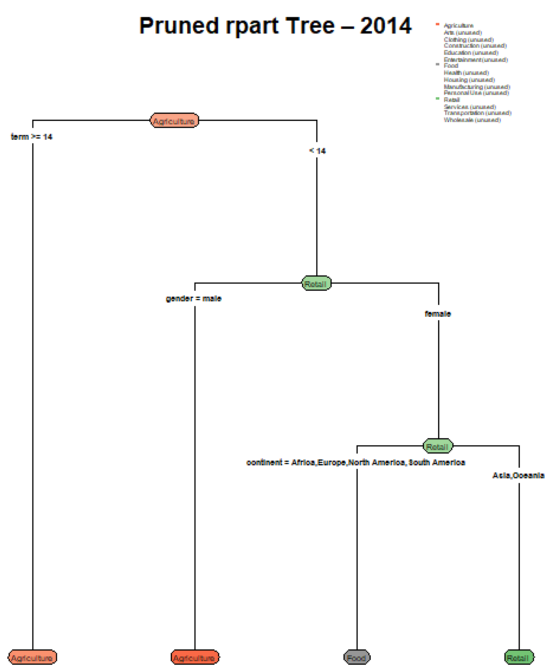
Leigh Lebos, Jessica . "How Microfinance Supports Livelihoods in Developing Countries." *Www.kiva.org*, 6 Sept. 2022, www.kiva.org/blog/how-microfinance-supports-livelihoods-in-developing-countries.

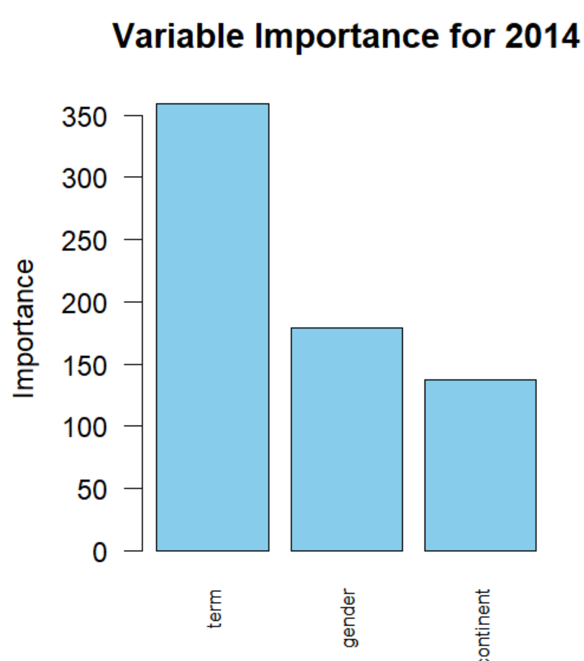
Appendix

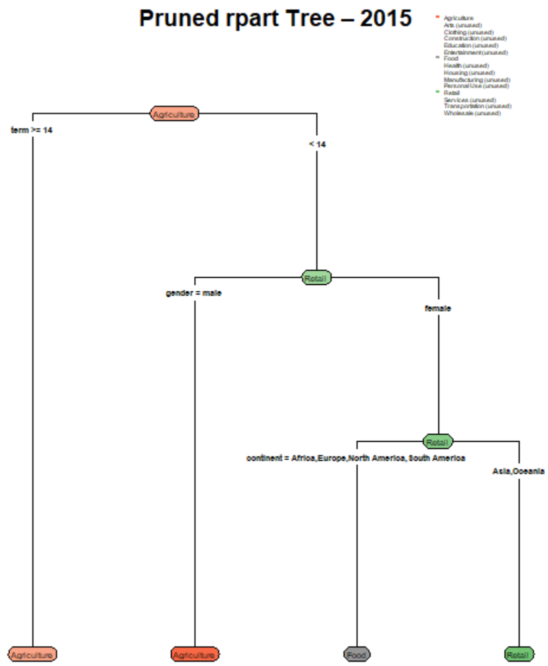
Appendix A: Sector Distribution for Each Year

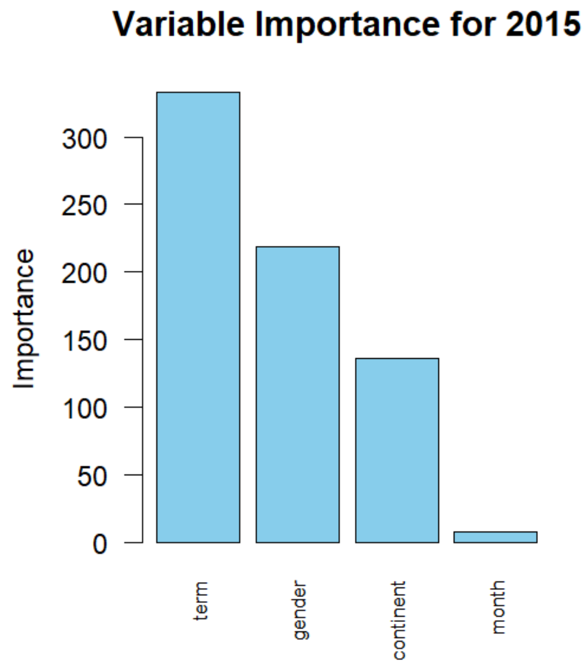


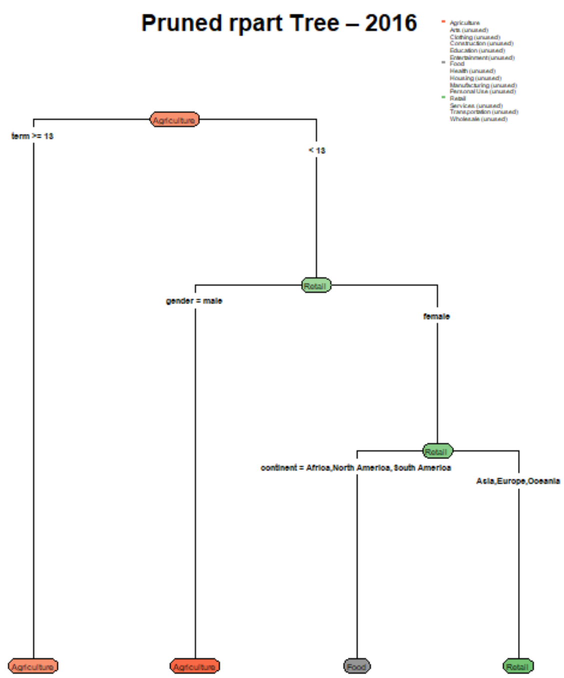
Appendix B: Tree Results from ctree

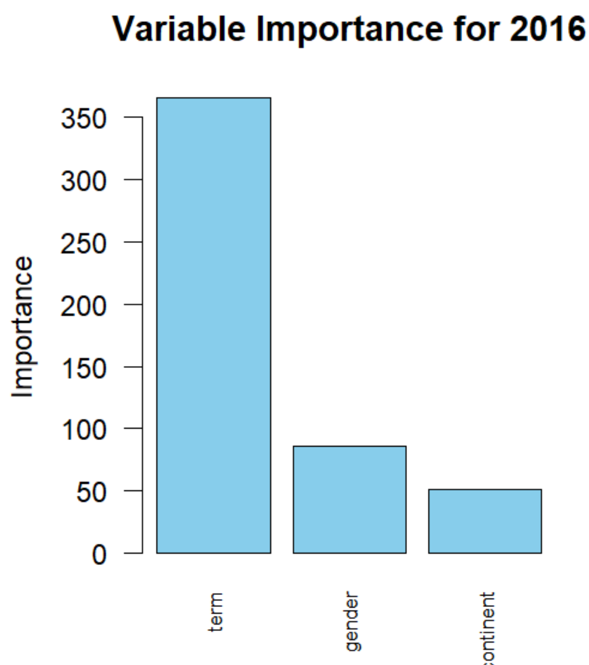


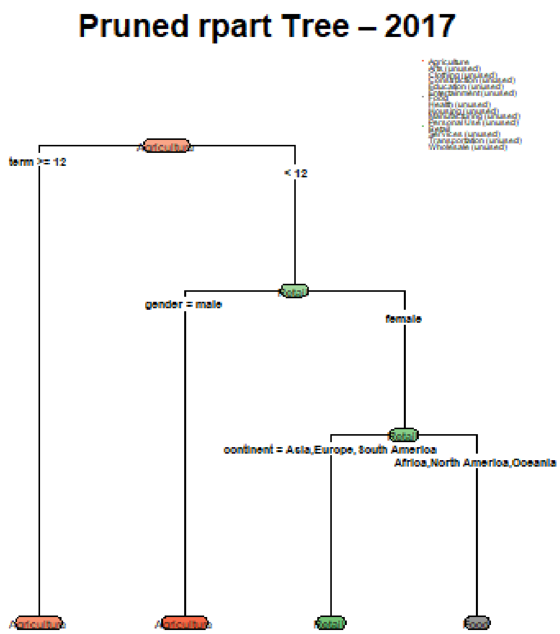


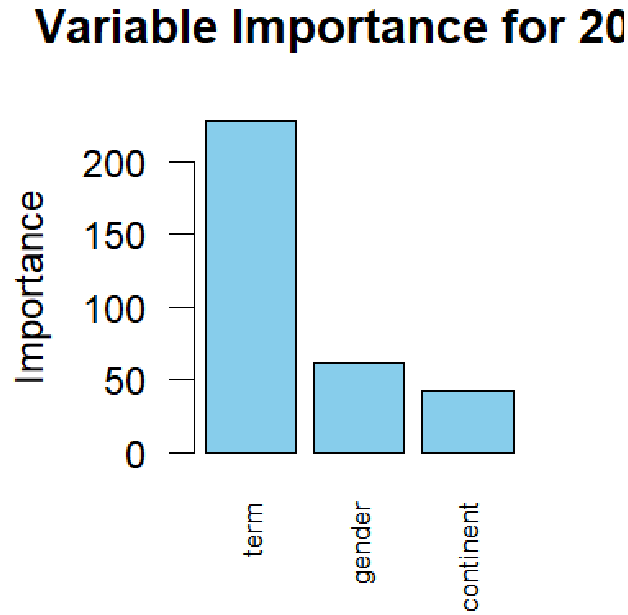

Pruned rpart Tree – 2015
Agriculture
term >= 14
< 14
Retail
gender = male
female
Retail
continent = Africa,Europe,North America,South America
Asia,Oceania
Agriculture
Agriculture
Food
Retail


Variable Importance for 2015
Importance
0
50
100
150
200
250
300
term
gender
continent
month

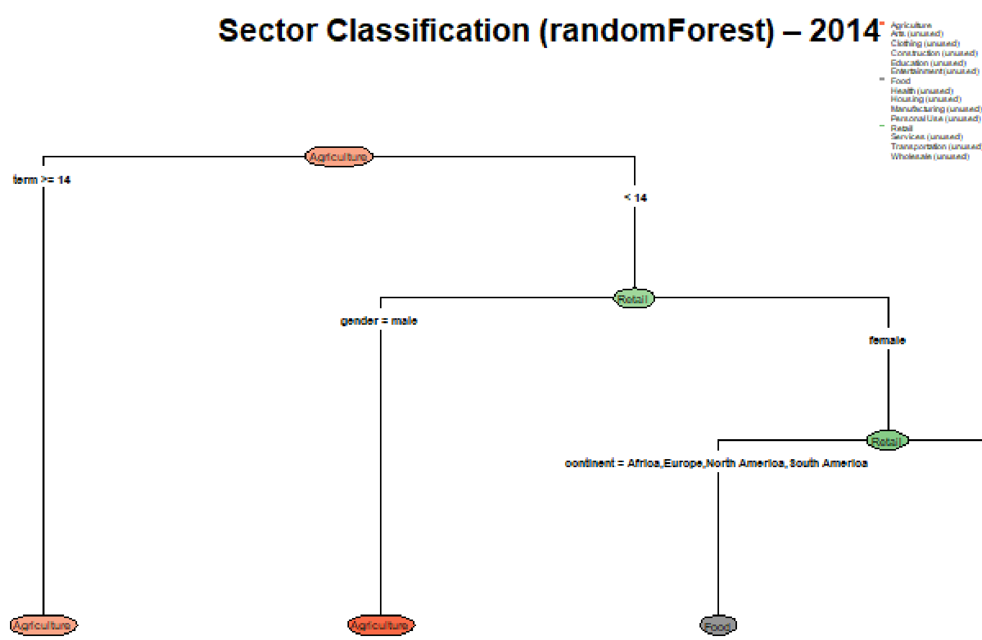

Pruned rpart Tree – 2016
Agriculture
term >= 13
< 13
Retail
gender = male
female
Retail
continent = Africa,North America,South America
Asia,Europe,Oceania
Agriculture
Agriculture
Food
Retail


Variable Importance for 2016
Importance
0
50
100
150
200
250
300
350
term
gender
continent

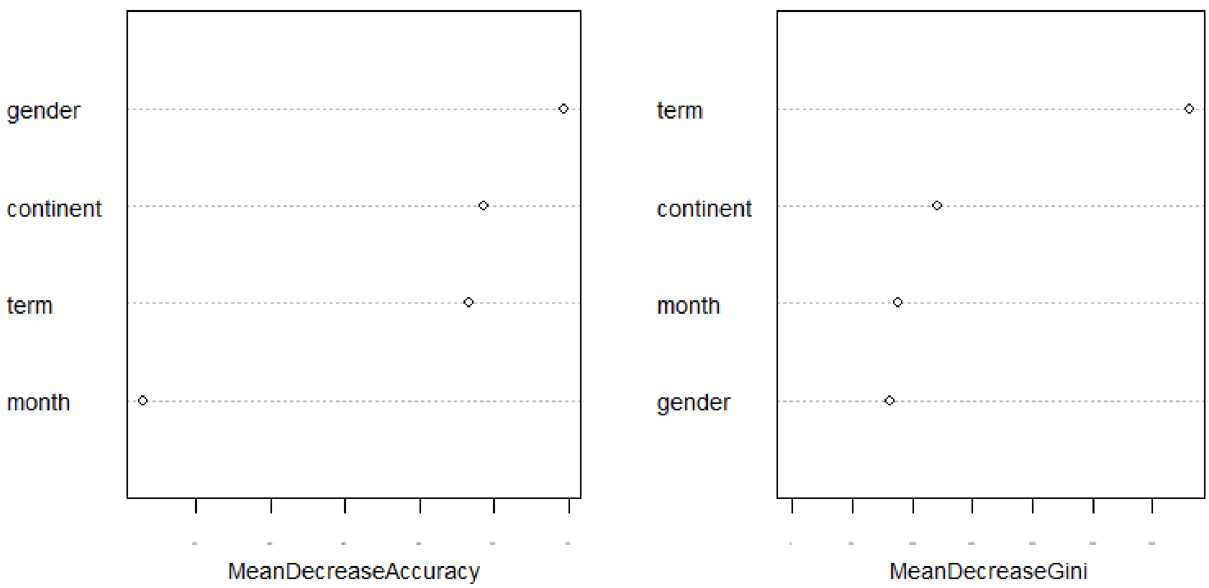




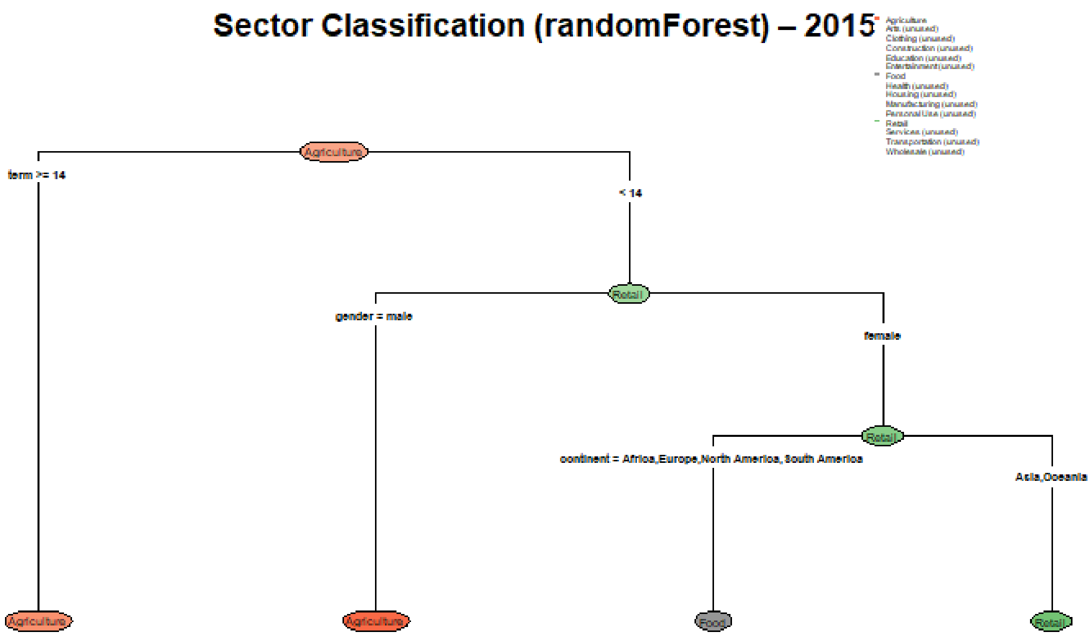
Appendix C: Tree Results from randomForest

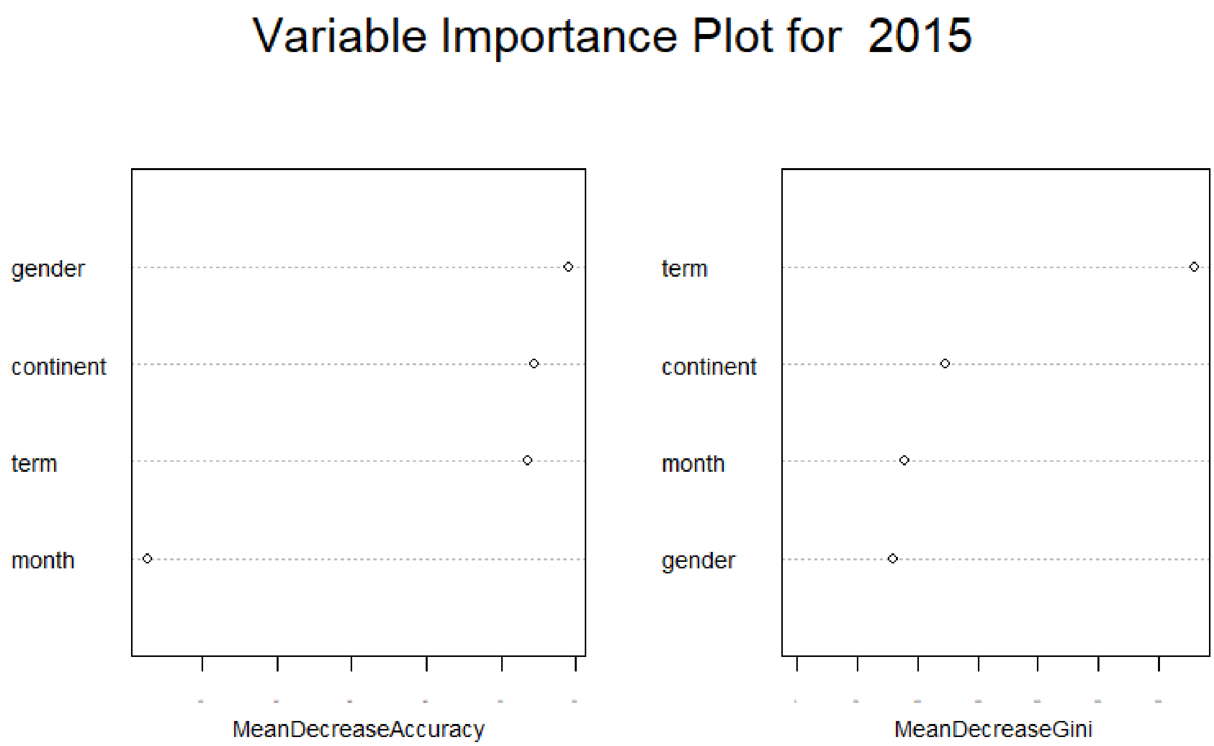


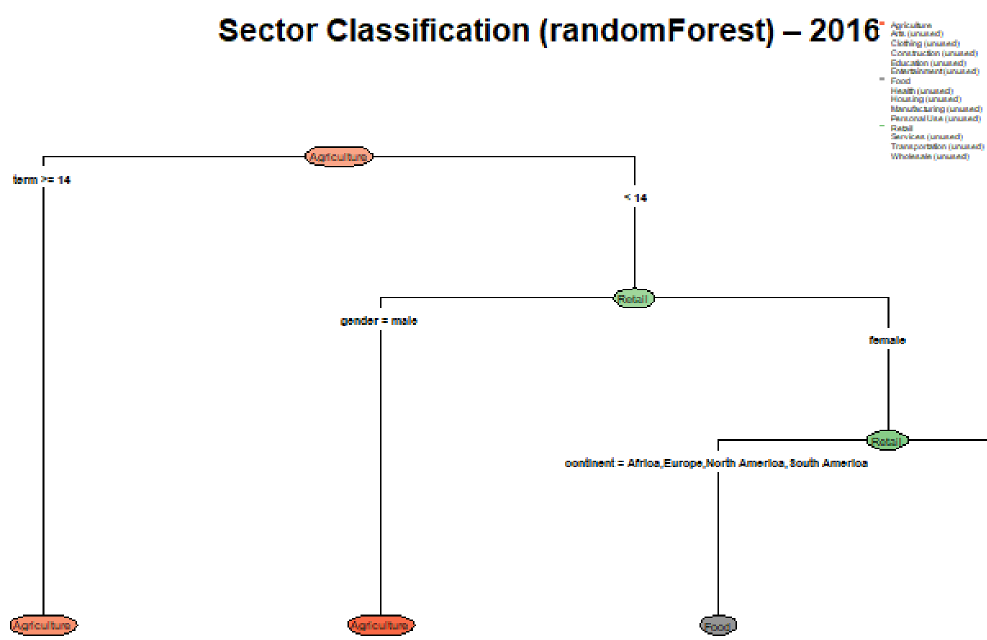
Variable Importance Plot for 2014

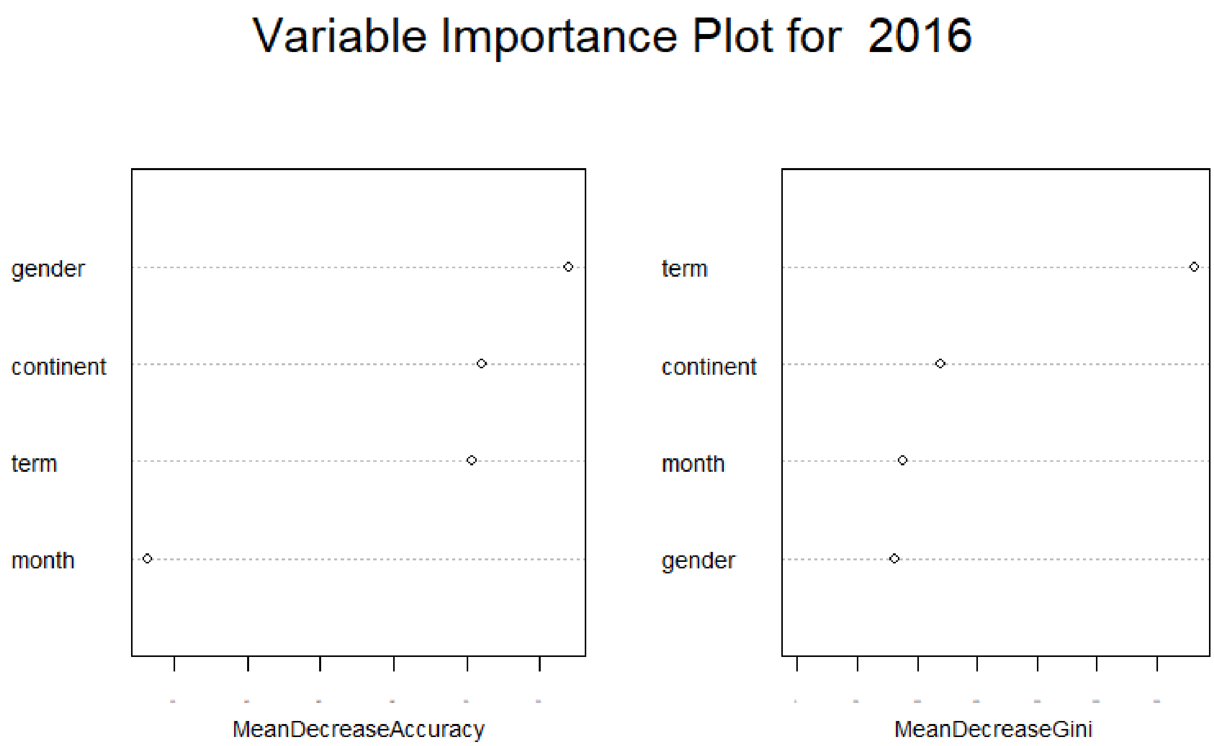


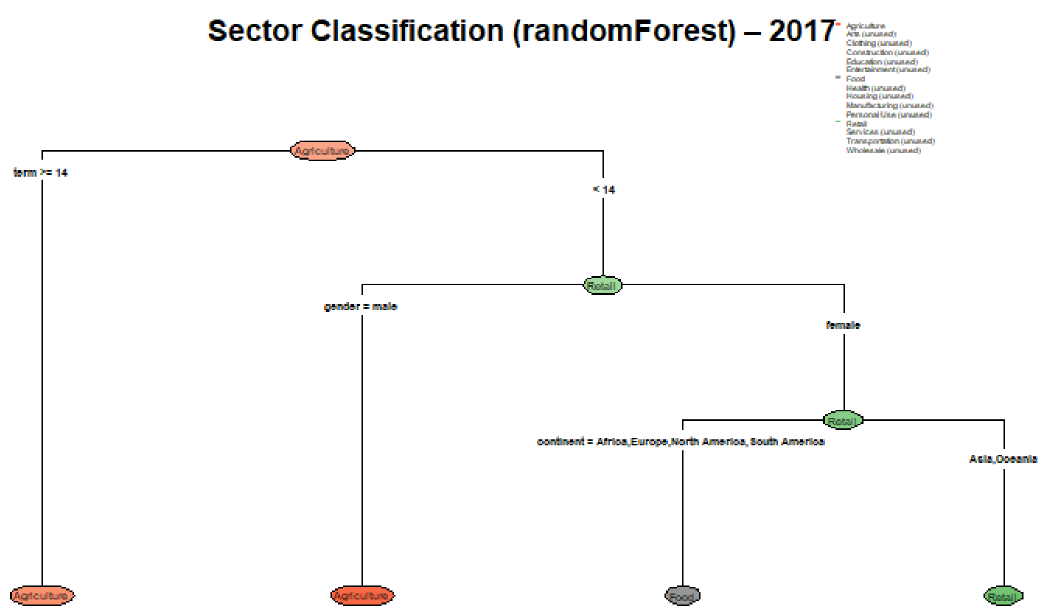
Sector Classification (randomForest) – 2015

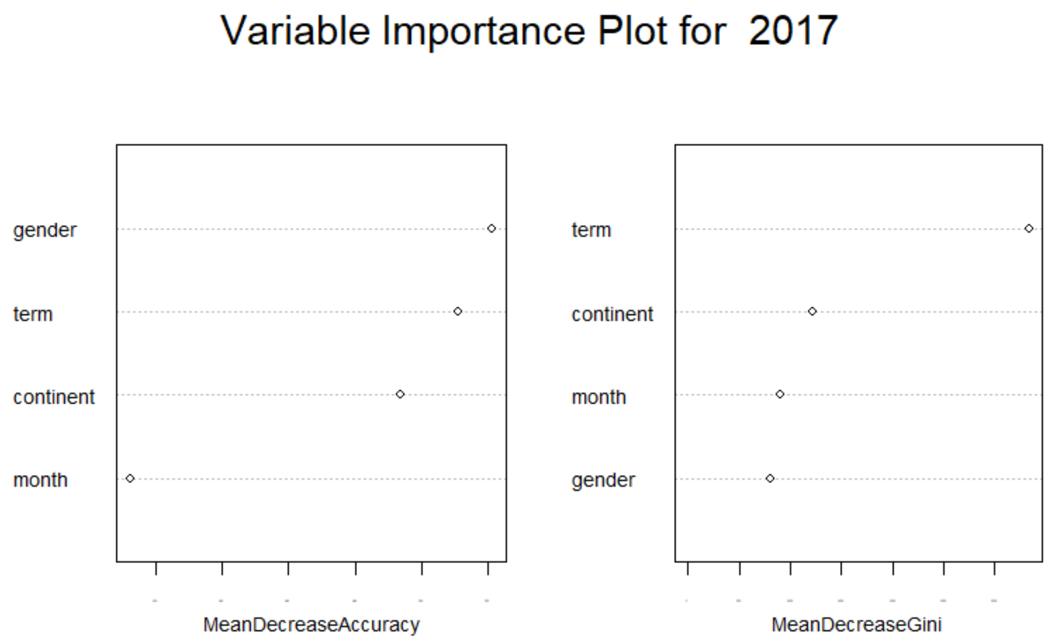



Variable Importance Plot for 2015
gender
continent
term
month
MeanDecreaseAccuracy
term
continent
month
gender
MeanDecreaseGini


Sector Classification (randomForest) – 2016
Agriculture
Arts (unused)
Clothing (unused)
Construction (unused)
Education (unused)
Entertainment (unused)
Food
Health (unused)
Housing (unused)
Manufacturing (unused)
Personal Use (unused)
Retail
Services (unused)
Transportation (unused)
Wholesale (unused)
Agriculture
term >= 14
< 14
Retail
gender = male
female
Retail
continent = Africa,Europe,North America,South America
Asia,Oceania
Agriculture
Agriculture
Food
Retail


Variable Importance Plot for 2016
gender
continent
term
month
MeanDecreaseAccuracy
term
continent
month
gender
MeanDecreaseGini


Sector Classification (randomForest) – 2017
Agriculture
Arts (unused)
Clothing (unused)
Construction (unused)
Education (unused)
Entertainment (unused)
Food
Health (unused)
Housing (unused)
Manufacturing (unused)
Personal Use (unused)
Retail
Services (unused)
Transportation (unused)
Wholesale (unused)
Agriculture
term >= 14
< 14
Retail
gender = male
female
Retail
continent = Africa,Europe,North America,South America
Asia,Oceania
Agriculture
Agriculture
Food
Retail



Appendix D: Class Imbalance Table

Sector	Loan Frequency
Agriculture	28286
Food	21527
Retail	21134
Services	7713
Housing	6080
Education	5645
Clothing	4974
Personal Use	3289
Transportation	2900
Arts	1953
Health	1608
Construction	1109
Manufacturing	1034
Entertainment	150
Wholesale	98